

Willian Vieira PhD

Geospatial data scientist with a PhD in quantitative ecology. I build machine learning and remote-sensing pipelines on large-scale, noisy, real-world data — from statistical modelling and satellite imagery processing to the production infrastructure that runs them.

DATA SCIENCE & ENGINEER EXPERIENCE

2025
|
2024
(1 yr 9 m)

- Data Analyst**
Habitat, Montreal, Canada
Architected end-to-end R/Python ETL and geospatial ML pipelines to process satellite imagery and large-scale spatial datasets for production analytics | Designed a geospatial lakehouse on Azure for streaming heterogeneous datasets from cloud storage and optimizing feature retrieval for ML training and inference | Built internal API layers and metadata-driven workflows to track provenance, validation, and dataset reliability | Pushed geospatial ML models into production, including land-use layers behind client-facing products | Led the transition to a Unix-based production environment using Docker, CI/CD, automated setup scripts, and DevOps best practices

2024
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2017

- PhD Research**
Integrative Ecology Lab, Sherbrooke, Canada
Built Bayesian hierarchical models to forecast forest dynamics from noisy, biased, continental-scale climate and field-inventory datasets | Developed reproducible HPC workflows (🔗) to run thousands of simulations in parallel for model evaluation, uncertainty propagation, and large-scale inference | Built end-to-end geospatial data pipelines to ingest, clean, harmonize, and model heterogeneous spatial datasets across North America | Developed open-source software libraries (🔗, 🔗) for spatial demographic modelling with version control, automated testing, and reproducible workflows | Published a **technical methods book** documenting the full computational pipeline, along with one peer-reviewed **publication** and two preprints (📄, 📄)

2022
|
2020
(part-time)

- Biostatistician**
Environment and Climate Change Canada - Quebec, Canada
Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible **geospatial pipeline** with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

EDUCATION

2024
|
2017

- PhD, Ecology**
Université de Sherbrooke - Sherbrooke, Canada
📄 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016
|
2015

- Masters 2, Agroecology and Resource Management**
Bordeaux Sciences Agro, Bordeaux, France
📄 *Modelling the dispersion of weed species in agricultural landscapes*

2015
|
2010

- BSc in Agronomy**
Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONTACT

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in [Linkedin/WillVieira](#)
🐙 [GitHub/WillVieira](#)
📄 Publications

SKILLS

Machine Learning & Statistics:
Geospatial ML · Remote sensing · Bayesian inference · Hierarchical models · Forecasting · Uncertainty · Model evaluation · Feature engineering

Programming & Database:
Python · R · SQL · PyTorch · Stan · Julia · DuckDB · Bash

Data Engineering & Cloud:
ETL/ELT · Azure · AWS · Docker · STAC · COG · GeoParquet · Apache Arrow · HPC

DevOps & Reproducibility:
Git · CI/CD · Unix · Unit testing · Make · Quarto · Jupyter · Automated validation

OUTREACH

· Developed and maintained the **QCBS R Workshop Series**, reaching nearly one thousand graduate students · Taught programming, statistics, and ecology to 250+ students across biology, engineering, and graduate programs · Published **5 papers & preprints** with ~100 citations · Translated stakeholder needs into technical analyses, reproducible workflows, and clear analytical outputs

LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional

CV source code hosted on 🔗
Last updated: 2026-05-11
📄 PDF download available